Lia: Review Packet

1.1–1.6 and 5.1–5.3

Part A: Student pages (pages 2–20)

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Part B: Answer keys, tutor only (pages 21–28)

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To print just Lia's copy, print pages 2–20. For your copy, print 21–28.

Worksheet: 1.1–1.6 and 5.1–5.3

Name: _____ Date: _____

One Easy (E) and one Hard (H) problem per section.

1.1 Change in tandem

1E. Use the table. Is f increasing or decreasing? Concave up or concave down? Justify with the changes.

x	0	1	2	3	4
$f(x)$	10	6	3	1	0

1H. A function is **decreasing and concave down** on $(0, 3)$, then **decreasing and concave up** on $(3, 6)$.

- (a) Sketch a graph that does this.
- (b) What is special about $x = 3$, and what is the rate of change doing there?
- (c) Give a real-world situation that behaves like this.

1.2 Rates of change

2E. Find the average rate of change of $f(x) = x^2 + 3x$ on $[1, 4]$.

2H. A car's position is $s(t) = t^3$ feet after t seconds.

- (a) Find the average velocity on $[2, 2.1]$ and on $[1.9, 2]$. (Calculator OK)
- (b) Use them to estimate the velocity at exactly $t = 2$. Include units.

1.3 Linear and quadratic functions

3E. Linear, quadratic or neither? Write the function.

x	0	1	2	3
y	5	8	11	14

3H. A quadratic f has $f(0) = 3$, $f(1) = 6$, $f(2) = 13$.

- (a) Use first and second differences to predict $f(3)$ and $f(4)$.
- (b) Find the formula $f(x) = ax^2 + bx + c$ and check your predictions.

1.4 Polynomials and rates of change

4E. $f(x) = (x+3)(x-1)(x-4)$. List the zeros. Is f positive or negative on $(1, 4)$?

4H. Write a degree-4 polynomial with a **double zero at** $x = -1$, a **double zero at** $x = 3$, and $f(0) = 18$.

- (a) Find it in factored form.
- (b) What does the graph do at each zero? What is the end behavior?
- (c) What is the most turning points it can have?

1.5 Complex zeros, multiplicity, symmetry

5E. Solve $x^2 - 6x + 13 = 0$.

5H. $f(x) = x^3 + 2x^2 + 9x + 18$.

- (a) Find **all** zeros, real and complex. (Hint: grouping.)
- (b) How many times does the graph touch the x-axis? Why so few?

1.6 End behavior

6E. Describe the end behavior of $f(x) = -3x^4 + x$ in words and with limit notation.

6H. $f(x) = -2(x-1)^3(x+2)^2$.

- (a) Find the leading term without multiplying everything out, and give the end behavior.
- (b) Describe the graph at each zero, and find the y-intercept.
- (c) Sketch it.

5.1 Composition

7E. $f(x) = 3x - 1$ and $g(x) = x^2 + 2$. Find $f(g(2))$ and $g(f(2))$.

7H. $f(x) = \sqrt{x-1}$ and $g(x) = \dfrac{1}{x-3}$.

- (a) Find $f(g(x))$.
- (b) Find its domain. (Both steps must work: g has to be defined, and $g(x)$ has to be allowed inside f.)

5.2 Inverses

8E. Find the inverse of $f(x) = 4x + 7$, and check that $f(f^{-1}(x)) = x$.

8H. $f(x) = \dfrac{2x-1}{x+3}$.

- (a) Find $f^{-1}(x)$.
- (b) Give the domain of f and of f^{-1}.
- (c) $f(1) = \frac{1}{4}$. Use your inverse to confirm $f^{-1}\left(\frac{1}{4}\right) = 1$.

5.3 Exponential functions

9E. Find the exponential function $y = ab^x$ through $(0, 5)$ and $(1, 15)$. Is it growth or decay, and by what percent?

9H.

- (a) Solve $4^{x+1} = 8^x$ by writing both sides with the same base.
- (b) A bacteria culture starts with 500 and **triples every 4 hours**. Write a model, find the population after 10 hours (calculator OK), and find exactly when it reaches 4500.

Practice Test: AP Precalc 1.1–1.6

Name: _____ Date: _____

About 45 minutes. No calculator unless marked.

1. (1.1) Use the table.

x	-2	-1	0	1	2	3
$f(x)$	8	3	0	-1	0	3

- (a) On what interval is f decreasing? Increasing?
- (b) Where are the zeros?
- (c) Is f concave up or concave down? Justify using the table.

2. (1.2) Find the average rate of change of $f(x) = x^3$ on $[1, 3]$.

3. (1.2) The temperature is $50°$F at 2 PM and $62°$F at 6 PM. Find the average rate of change with units, and say what it means.

4. (1.2) Estimate the rate of change of $g(x) = x^2 + 1$ at $x = 2$ using the intervals $[2, 2.1]$ and $[1.9, 2]$. (Calculator OK)

5. (1.3) Linear, quadratic or neither? Justify.

x	0	1	2	3	4
(a)	1	4	7	10	13
(b)	3	4	7	12	19
(c)	1	2	4	8	16

6. (1.3) $f(x) = 2x^2$. Find the average rate of change on $[0, 1]$, $[1, 2]$ and $[2, 3]$. By how much does it change each time?

7. (1.4) $f(x) = x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$.

- (a) What are the real zeros?
- (b) Is f positive or negative on $(1, 2)$? On $(2, 3)$?
- (c) What is the most turning points f could have?

8. (1.4) A polynomial has degree 5. What is the most real zeros it can have? The most turning points? The fewest real zeros?

9. (1.5) Find all zeros of $f(x) = x^4 - 16$.

10. (1.5) Solve $x^2 + 2x + 5 = 0$.

11. (1.5) Write a polynomial of least degree, with real coefficients and leading coefficient 1, that has a zero of 3 with multiplicity 2 and a zero of i. Give it in standard form.

12. (1.5) For $f(x) = x^2(x-4)^3$, list each zero with its multiplicity and say whether the graph crosses or bounces there.

13. (1.5) Even, odd or neither? Show the test. (a) $x^5 - 2x^3$ (b) $x^4 + x^2 - 7$ (c) $x^2 + x$

14. (1.6) Describe the end behavior with limit notation. (a) $f(x) = 4x^6 - x^7 + 1$ (b) $g(x) = -(x-1)^2(x+3)^2$ (c) $h(x) = 2x(x-1)(x+2)^2$

15. (1.4–1.6) Sketch $f(x) = (x+1)^2(x-2)$. Label the zeros, the y-intercept and the end behavior.

16. (AP-style free response, 1.1–1.4) The water level $W(t)$, in inches, in a tank is measured every hour.

t (hours)	0	1	2	3	4	5
$W(t)$ (in.)	20	25	36	47	52	45

- (a) Find the average rate of change of W from $t = 1$ to $t = 4$. Include units and interpret it.
- (b) Over which one-hour interval(s) does the water level rise fastest?
- (c) Is W concave up or concave down on $[3, 5]$? Justify with the table.
- (d) Based on the table, around when does the water level reach a local maximum? Explain.

Practice Test: 5.1–5.3 (Composition, Inverses, Exponentials)

Name: _____ Date: _____

About 50 minutes. Calculator only where marked.

1. (5.1) $f(x) = x^2 - 1$ and $g(x) = 3x + 2$. Find:

- (a) $f(g(1))$ (b) $g(f(1))$ (c) $f(f(2))$
- (d) $f(g(x))$ (e) $g(f(x))$

2. (5.1) Use the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	2	1	4	3

Find $f(g(1))$, $g(f(1))$, $f(f(3))$ and $g(g(4))$.

3. (5.1) Find each composite and its domain.

- (a) $f(x) = \sqrt{x}$, $g(x) = x - 5$: find $f(g(x))$ and $g(f(x))$, each with its domain.
- (b) ★ $f(x) = \dfrac{2}{x-1}$, $g(x) = \dfrac{4}{x}$: find $f(g(x))$ and its domain.

4. (5.1) Find f and g so that $h(x) = f(g(x))$. (a) $h(x) = (2x+5)^3$ (b) $h(x) = \sqrt{x^2+1}$ (c) $h(x) = \dfrac{1}{x-7}$

5. (5.1) A company makes $n(t) = 25t$ shirts in t hours, and it costs $C(n) = 10n + 50$ dollars to make n shirts.

- (a) Find $C(n(t))$ and say what it represents, with units.
- (b) Find $C(n(4))$ and interpret it.

6. (5.2) One-to-one or not? Explain.

- (a) $\{(1,4),(2,5),(3,4)\}$ (b) $\{(-1,2),(0,3),(2,7)\}$
- (c) $y = x^2$ (d) $y = x^3$ (e) $y = |x|$ (f) $y = 2^x$

7. (5.2) f passes through $(-2, 1)$, $(0, 3)$ and $(4, 7)$, has domain $[-2, 4]$ and range $[1, 7]$.

- (a) List three points on f^{-1}.
- (b) If $f(5) = 11$, what is $f^{-1}(11)$?
- (c) What are the domain and range of f^{-1}?

8. (5.2) Use the table: find $f^{-1}(11)$, $f^{-1}(5)$ and $f^{-1}(f(3))$.

x	0	1	2	3
$f(x)$	5	8	11	14

9. (5.2) Find the inverse.

- (a) $f(x) = 5x - 3$
- (b) $f(x) = x^3 + 4$
- (c) ★ $f(x) = \dfrac{x+1}{x-2}$. Also give the domain and range of f.

10. (5.2) Show that $f(x) = 2x - 6$ and $g(x) = \frac{x}{2} + 3$ are inverses.

11. (5.2) ★ $f(x) = (x-2)^2$ is not one-to-one.

- (a) Restrict its domain so it is, keeping the right half.
- (b) Find f^{-1} and give its domain and range.

12. (5.2) The graph of f contains the points $(0, 1)$, $(1, 2)$ and $(2, 4)$. Sketch f and f^{-1} on the same axes along with the line $y = x$. Where do the points go?

13. (5.3) $f(x) = 3 \cdot 2^x$. Find $f(0)$, $f(3)$ and $f(-1)$.

14. (5.3) For each, give the initial value, growth or decay, and the percent rate. (a) $500(1.03)^t$ (b) $80(0.6)^t$

15. (5.3) $f(x) = 2^{x-1} + 3$. Describe the transformation from 2^x, and give the domain, range, horizontal asymptote and y-intercept.

16. (5.3) Find the exponential function $y = ab^x$ through each pair of points. (a) $(0, 4)$ and $(2, 36)$ (b) $(1, 10)$ and $(3, 2.5)$

17. (5.3) Solve. (a) $2^{x+1} = 16$ (b) $9^x = 27$ (c) $\left(\frac{1}{2}\right)^x = 8$ (d) ★ $3^{x^2} = 81$

18. (5.3) \$5,000 is invested at 6% for 5 years. Find the balance compounded quarterly and compounded continuously. (Calculator)

19. (5.3) A 200 g sample has a half-life of 10 years. Write a model and find how much is left after 25 years. (Calculator)

20. (5.1–5.3, putting it together) $f(x) = 2^x$ and $g(x) = x - 3$.

- (a) Find $h(x) = f(g(x))$. Describe it as a transformation of 2^x.
- (b) Is h one-to-one? How do you know?
- (c) $h(5) = 4$. What point must be on the graph of h^{-1}?
- (d) What is the horizontal asymptote of h? What vertical line would the graph of h^{-1} approach instead?

Answer Key

1E. Changes: $-4, -3, -2, -1$. All negative, so **decreasing**. The changes are increasing (getting less negative), so the rate of change is increasing: **concave up**.

1H. (a) Falling and curving down, then still falling but flattening out. (b) $x = 3$ is a **point of inflection**. The rate of change is decreasing up to $x = 3$ and increasing after, so the graph is falling **fastest** at $x = 3$ (the most negative rate). (c) A sled going down a hill that's steepest in the middle; or a tank draining faster and faster, then easing off.

2E. $f(4) = 28$, $f(1) = 4$, so $\dfrac{28-4}{4-1} = \boxed{8}$.

2H. (a) $\dfrac{2.1^3 - 8}{0.1} = \dfrac{9.261 - 8}{0.1} = 12.61$ ft/s and $\dfrac{8 - 1.9^3}{0.1} = \dfrac{8 - 6.859}{0.1} = 11.41$ ft/s. (b) The true value is between them, about **12 ft/s** (their average is 12.01).

3E. Differences all 3, so **linear**: $y = 3x + 5$.

3H. (a) First differences $3, 7$; second difference 4. Continue: next first differences $11, 15$, so $f(3) = 13 + 11 = 24$ and $f(4) = 24 + 15 = 39$. (b) $c = 3$; $a + b = 3$; $4a + 2b = 10$, so $a = 2$, $b = 1$: $f(x) = 2x^2 + x + 3$. Check: $f(3) = 18 + 3 + 3 = 24$ ✓, $f(4) = 32 + 4 + 3 = 39$ ✓.

4E. Zeros $-3, 1, 4$. Test $x = 2$: $(5)(1)(-2) = -10$, so **negative** on $(1, 4)$.

4H. (a) $f(x) = a(x+1)^2(x-3)^2$ and $f(0) = a(1)(9) = 9a = 18$, so $a = 2$: $\boxed{f(x) = 2(x+1)^2(x-3)^2}$. (b) Both zeros have even multiplicity, so the graph **bounces** at -1 and at 3. The leading term is $2x^4$: up on both ends. (c) At most $4 - 1 = 3$ turning points (it has exactly 3: min, max, min).

5E. $x = \dfrac{6 \pm \sqrt{36 - 52}}{2} = \dfrac{6 \pm 4i}{2} = \boxed{3 \pm 2i}$

5H. (a) $x^2(x+2) + 9(x+2) = (x+2)(x^2+9)$, so $x = -2, \pm 3i$. (b) Only **once**, at $x = -2$. The other two zeros are complex, so the graph never reaches them.

6E. Leading term $-3x^4$: even degree, negative, so **down on both ends**. $\lim_{x \to \pm\infty} f(x) = -\infty$.

6H. (a) $-2 \cdot x^3 \cdot x^2 = -2x^5$: odd degree, negative. $\lim_{x \to -\infty} f = \infty$ and $\lim_{x \to \infty} f = -\infty$. (b) At $x = 1$, multiplicity 3: **crosses**, flattening like x^3. At $x = -2$, multiplicity 2: **bounces**. y-intercept: $-2(-1)^3(2)^2 = -2(-1)(4) = 8$. (c) Comes down from the top left, bounces off -2, rises through $(0, 8)$, turns, flattens through 1, then heads down.

7E. $g(2) = 6$, so $f(6) = 17$. $f(2) = 5$, so $g(5) = 27$.

7H. (a) $f(g(x)) = \sqrt{\dfrac{1}{x-3} - 1}$. (b) g needs $x \neq 3$. f needs its input ≥ 1: $\dfrac{1}{x-3} \geq 1$. That only works when $x - 3$ is positive and at most 1, so $0 < x - 3 \leq 1$. Domain: $\boxed{3 < x \leq 4}$.

8E. $x = 4y + 7$, so $y = \dfrac{x-7}{4}$. Check: $4 \cdot \dfrac{x-7}{4} + 7 = x$ ✓.

8H. (a) Swap: $x = \dfrac{2y-1}{y+3}$, so $xy + 3x = 2y - 1$, then $xy - 2y = -3x - 1$, then $y(x-2) = -3x - 1$. So $\boxed{f^{-1}(x) = \dfrac{3x+1}{2-x}}$. (b) Domain of f: $x \neq -3$. Domain of f^{-1}: $x \neq 2$ (which is the range of f). (c) $f^{-1}\left(\frac{1}{4}\right) = \dfrac{3/4 + 1}{2 - 1/4} = \dfrac{7/4}{7/4} = 1$ ✓.

9E. $a = 5$; $5b = 15$, so $b = 3$: $y = 5 \cdot 3^x$. Growth, $b - 1 = 2$, so **200%** per step.

9H. (a) $2^{2(x+1)} = 2^{3x}$, so $2x + 2 = 3x$ and $\boxed{x = 2}$. (b) $N(t) = 500 \cdot 3^{t/4}$. $N(10) = 500 \cdot 3^{2.5} \approx \boxed{7794}$. $500 \cdot 3^{t/4} = 4500$ gives $3^{t/4} = 9 = 3^2$, so $t/4 = 2$ and $\boxed{t = 8 \text{ hours}}$.

Worked Answer Key: Practice Test 1.1–1.6

1. (1.1) Table: $f = 8, 3, 0, -1, 0, 3$ at $x = -2, \ldots, 3$.

- First differences (each step of 1): $3 - 8 = -5$, $0 - 3 = -3$, $-1 - 0 = -1$, $0 - (-1) = 1$, $3 - 0 = 3$.
- (a) Negative differences until $x = 1$, then positive. **Decreasing on $(-2, 1)$, increasing on $(1, 3)$.** The lowest point in the table is $(1, -1)$.
- (b) Zeros are where $f(x) = 0$: $\boxed{x = 0 \text{ and } x = 2}$.
- (c) The differences $-5, -3, -1, 1, 3$ are increasing (the rate of change is increasing), so f is **concave up**. Bonus: the second differences are all 2, so it's quadratic: $f(x) = x^2 - 2x$.

2. (1.2) $f(x) = x^3$ on $[1, 3]$.

- $f(3) = 27$ and $f(1) = 1$.
- $\dfrac{27 - 1}{3 - 1} = \dfrac{26}{2} = \boxed{13}$

3. (1.2) 50°F at 2 PM, 62°F at 6 PM.

- $\dfrac{62 - 50}{6 - 2} = \dfrac{12}{4} = \boxed{3 \text{ °F per hour}}$
- Meaning: between 2 and 6 PM the temperature rose an **average** of 3 degrees each hour. It may not have risen exactly 3 degrees in every single hour.

4. (1.2) Rate of change of $g(x) = x^2 + 1$ at $x = 2$.

- $g(2) = 5$, $g(2.1) = 4.41 + 1 = 5.41$, $g(1.9) = 3.61 + 1 = 4.61$.
- Right side: $\dfrac{5.41 - 5}{2.1 - 2} = \dfrac{0.41}{0.1} = 4.1$
- Left side: $\dfrac{5 - 4.61}{2 - 1.9} = \dfrac{0.39}{0.1} = 3.9$
- The true rate is between them: $\boxed{\approx 4}$. (Smaller intervals, like $[2, 2.01]$, would close in further.)

5. (1.3) Linear, quadratic or neither?

- (a) $1, 4, 7, 10, 13$: first differences $3, 3, 3, 3$, constant. **Linear.**
- (b) $3, 4, 7, 12, 19$: first differences $1, 3, 5, 7$; second differences $2, 2, 2$, constant. **Quadratic.**
- (c) $1, 2, 4, 8, 16$: first differences $1, 2, 4, 8$; second differences $1, 2, 4$, not constant. **Neither.** (The outputs double each time: constant ratio 2, so it's exponential.)

6. (1.3) $f(x) = 2x^2$.

- Values: $f(0) = 0$, $f(1) = 2$, $f(2) = 8$, $f(3) = 18$.
- Rates: $[0, 1]$: $\frac{2-0}{1} = 2$. $[1, 2]$: $\frac{8-2}{1} = 6$. $[2, 3]$: $\frac{18-8}{1} = 10$.
- They go up by **4** each time. For a quadratic the average rate of change changes at a constant rate. That's the 1.3 fact being tested.

7. (1.4) $f(x) = (x-1)(x-2)(x-3)$.

- (a) Zeros: $\boxed{1, 2, 3}$.
- (b) Test a point in each interval:
  - $(1, 2)$, test $x = 1.5$: $(0.5)(-0.5)(-1.5) = +0.375$, so **positive**.
  - $(2, 3)$, test $x = 2.5$: $(1.5)(0.5)(-0.5) = -0.375$, so **negative**.
- (c) Degree 3, so at most $3 - 1 = \boxed{2}$ turning points.

8. (1.4) Degree 5.

- At most **5** real zeros (the degree).
- At most **4** turning points (degree − 1).
- At least **1** real zero. Odd degree means the two ends go in opposite directions (one up, one down), so the graph has to cross the x-axis at least once. (Also, complex zeros come in pairs, so 5 zeros can't all be complex.)

9. (1.5) Zeros of $x^4 - 16$.

- Difference of squares: $(x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$.
- Again: $x^2 - 4 = (x - 2)(x + 2)$.
- $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.
- Zeros: $\boxed{2,\ -2,\ 2i,\ -2i}$ (4 zeros for degree 4 ✓).

10. (1.5) $x^2 + 2x + 5 = 0$.

- Doesn't factor, so use the quadratic formula with $a = 1$, $b = 2$, $c = 5$.
- Discriminant: $b^2 - 4ac = 4 - 20 = -16$. Negative means complex zeros.
- $x = \dfrac{-2 \pm \sqrt{-16}}{2} = \dfrac{-2 \pm 4i}{2} = \boxed{-1 \pm 2i}$

11. (1.5) Least degree, zero 3 (multiplicity 2) and zero i.

- Real coefficients mean $-i$ must also be a zero. Zeros: $3, 3, i, -i$, so degree 4.
- $(x - 3)^2 = x^2 - 6x + 9$ and $(x - i)(x + i) = x^2 - i^2 = x^2 + 1$.
- Multiply: $(x^2 - 6x + 9)(x^2 + 1) = x^4 + x^2 - 6x^3 - 6x + 9x^2 + 9$.
- Combine: $\boxed{x^4 - 6x^3 + 10x^2 - 6x + 9}$

12. (1.5) $f(x) = x^2(x - 4)^3$.

- $x = 0$: factor x^2, multiplicity **2** (even), so the graph **bounces**.
- $x = 4$: factor $(x - 4)^3$, multiplicity **3** (odd), so it **crosses**, flattening out as it passes through (like x^3 at the origin).

13. (1.5) Plug in $-x$ and compare.

- (a) $(-x)^5 - 2(-x)^3 = -x^5 + 2x^3 = -(x^5 - 2x^3) = -f(x)$, so **odd**.
- (b) $(-x)^4 + (-x)^2 - 7 = x^4 + x^2 - 7 = f(x)$, so **even**.
- (c) $(-x)^2 + (-x) = x^2 - x$. That's not $f(x) = x^2 + x$, and not $-f(x) = -x^2 - x$, so **neither**.
- Shortcut: all even powers (a constant counts as x^0) means even, and all odd powers means odd. A mix means neither.

14. (1.6) End behavior.

- (a) Reorder: $-x^7 + 4x^6 + 1$. The leading term is $-x^7$: odd degree, negative. $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = -\infty$.
- (b) Leading term $-(x^2)(x^2) = -x^4$: even degree, negative. $\lim_{x \to \pm\infty} g(x) = -\infty$ (down on both ends).
- (c) Leading term $2x \cdot x \cdot x^2 = 2x^4$: even degree, positive. $\lim_{x \to \pm\infty} h(x) = \infty$ (up on both ends).

15. (1.4–1.6) Sketch $f(x) = (x+1)^2(x-2)$.

- Zeros: $x = -1$ (multiplicity 2, **bounce**) and $x = 2$ (multiplicity 1, **cross**).
- y-intercept: $f(0) = (1)^2(-2) = -2$, so $(0, -2)$.
- Leading term $x^2 \cdot x = x^3$: degree 3, positive, so down on the left and up on the right.
- Putting it together: start at the bottom left, rise up to touch the x-axis at -1 and turn back down, pass through $(0, -2)$, turn around (somewhere between 0 and 2), cross up through $x = 2$, and keep rising.

16. (AP-style) Water level $W(t)$: $20, 25, 36, 47, 52, 45$ at $t = 0, \ldots, 5$.

- (a) $\dfrac{W(4) - W(1)}{4 - 1} = \dfrac{52 - 25}{3} = \boxed{9 \text{ inches per hour}}$. Between hour 1 and hour 4, the water level rose an average of 9 inches each hour.
- (b) One-hour changes: $+5, +11, +11, +5, -7$. The biggest rise is **11 in/hr**, on $[1, 2]$ and $[2, 3]$.
- (c) On $[3, 5]$ the changes are $+5$ then -7. The rate of change is **decreasing**, so W is **concave down** there.
- (d) The level rises until $t = 4$ (change $+5$ on $[3, 4]$) and then falls (change -7 on $[4, 5]$). It switches from increasing to decreasing **at about** $t = 4$, so that's the local maximum (about 52 inches).
- AP grading tip: each part needs the number, the units, and a reason that cites the table.

Worked Answer Key: Practice Test 5.1–5.3

1. (5.1) $f(x) = x^2 - 1$, $g(x) = 3x + 2$.

- (a) $f(g(1))$: inside first, $g(1) = 3 + 2 = 5$. Then $f(5) = 25 - 1 = \boxed{24}$.
- (b) $g(f(1))$: $f(1) = 1 - 1 = 0$. Then $g(0) = \boxed{2}$.
- (c) $f(f(2))$: $f(2) = 3$. Then $f(3) = 9 - 1 = \boxed{8}$.
- (d) Replace x in f with $(3x+2)$: $f(g(x)) = (3x+2)^2 - 1 = 9x^2 + 12x + 4 - 1 = \boxed{9x^2 + 12x + 3}$.
- (e) Replace x in g with $(x^2 - 1)$: $g(f(x)) = 3(x^2 - 1) + 2 = 3x^2 - 3 + 2 = \boxed{3x^2 - 1}$.
- Check (d) with (a): $9 + 12 + 3 = 24$ ✓.

2. (5.1) Table lookups. Find the inside value in the g (or f) row, then look up **that number** in the x row.

- $f(g(1))$: $g(1) = 2$, then $f(2) = \boxed{4}$.
- $g(f(1))$: $f(1) = 3$, then $g(3) = \boxed{4}$.
- $f(f(3))$: $f(3) = 1$, then $f(1) = \boxed{3}$.
- $g(g(4))$: $g(4) = 3$, then $g(3) = \boxed{4}$.

3. (5.1) Composites and domains.

- (a) $f(g(x)) = \sqrt{x - 5}$. The square root needs $x - 5 \geq 0$, so the domain is $\boxed{x \geq 5}$. $g(f(x)) = \sqrt{x} - 5$. Only $\sqrt{x}$ needs care: $\boxed{x \geq 0}$.
- (b) $f(g(x)) = \dfrac{2}{\frac{4}{x} - 1}$. Multiply the top and bottom by x: $\dfrac{2x}{4 - x}$. Domain, checked **before** simplifying:
  1. $g(x) = \frac{4}{x}$ needs $x \neq 0$.
  2. f can't take an input of 1: $\frac{4}{x} \neq 1$, so $x \neq 4$.

  Domain: $\boxed{\text{all reals except } 0 \text{ and } 4}$. The simplified $\frac{2x}{4-x}$ looks fine at $x = 0$, but the original isn't. That's why you find the domain first.

4. (5.1) Decompose: find the inside piece and the outside action.

- (a) Inside: $g(x) = 2x + 5$. Outside: "cube it," $f(x) = x^3$.
- (b) Inside: $g(x) = x^2 + 1$. Outside: "square root it," $f(x) = \sqrt{x}$.
- (c) Inside: $g(x) = x - 7$. Outside: "1 over it," $f(x) = \frac{1}{x}$.
- Check: plug g into f and you get h back. Other splits can also be correct.

5. (5.1) $n(t) = 25t$ shirts in t hours; $C(n) = 10n + 50$ dollars.

- (a) Plug $n(t)$ in for n: $C(n(t)) = 10(25t) + 50 = \boxed{250t + 50}$. Units: hours → shirts → dollars, so this is **the cost in dollars of t hours of production**.
- (b) $n(4) = 100$ shirts, so $C(100) = 1000 + 50 = \boxed{1050}$. Four hours of production (100 shirts) costs \$1,050. Check with (a): $250(4) + 50 = 1050$ ✓.

6. (5.2) One-to-one?

- (a) The output 4 comes from both $x = 1$ and $x = 3$. **Not one-to-one.**
- (b) Outputs 2, 3, 7 are all different. **One-to-one.**
- (c) $y = x^2$: $2^2 = (-2)^2 = 4$, so it fails the horizontal line test. **No.**
- (d) $y = x^3$: always increasing, so every horizontal line hits once. **Yes.**
- (e) $y = |x|$: $|3| = |-3|$. **No.**
- (f) $y = 2^x$: always increasing. **Yes.**

7. (5.2) f through $(-2, 1)$, $(0, 3)$, $(4, 7)$; domain $[-2, 4]$, range $[1, 7]$.

- (a) Swap each pair: $\boxed{(1, -2),\ (3, 0),\ (7, 4)}$.
- (b) $f(5) = 11$ means $f^{-1}(11) = \boxed{5}$ (input and output trade places).
- (c) Domain and range swap: f^{-1} has domain $\boxed{[1, 7]}$ and range $\boxed{[-2, 4]}$.

8. (5.2) Read the table backwards: find the output in the $f(x)$ row and read up to x.

- $f^{-1}(11)$: $f(2) = 11$, so $\boxed{2}$.
- $f^{-1}(5)$: $f(0) = 5$, so $\boxed{0}$.
- $f^{-1}(f(3))$: an inverse undoes the function, so this is just $\boxed{3}$. (Or: $f(3) = 14$ and $f^{-1}(14) = 3$.)

9. (5.2) Find the inverse: write $y =$, swap x and y, solve for y.

- (a) $x = 5y - 3$. Add 3: $x + 3 = 5y$. Divide: $\boxed{f^{-1}(x) = \dfrac{x+3}{5}}$.
- (b) $x = y^3 + 4$. Subtract 4: $y^3 = x - 4$. Cube root: $\boxed{f^{-1}(x) = \sqrt[3]{x-4}}$.
- (c) $x = \dfrac{y+1}{y-2}$.
  1. Multiply both sides by $(y-2)$: $x(y-2) = y + 1$.
  2. Distribute: $xy - 2x = y + 1$.
  3. Get the y-terms on one side: $xy - y = 2x + 1$.
  4. Factor out y: $y(x-1) = 2x + 1$.
  5. Divide: $\boxed{f^{-1}(x) = \dfrac{2x+1}{x-1}}$.

  Domain of f: $x \neq 2$. Range of f = domain of f^{-1}: $y \neq 1$. (It matches: the horizontal asymptote of f is $y = 1$.)

10. (5.2) $f(x) = 2x - 6$, $g(x) = \frac{x}{2} + 3$.

- $f(g(x)) = 2\left(\frac{x}{2} + 3\right) - 6 = x + 6 - 6 = x$ ✓
- $g(f(x)) = \dfrac{2x-6}{2} + 3 = x - 3 + 3 = x$ ✓
- Both give x, so they're inverses. (You need to check both directions.)

11. (5.2) $f(x) = (x-2)^2$.

- (a) The vertex is at $x = 2$, and the right half is always increasing: restrict to $\boxed{x \geq 2}$.
- (b) $y = (x-2)^2$. Swap: $x = (y-2)^2$. Square root: $\sqrt{x} = y - 2$ (take the **positive** root, because $y \geq 2$ on the restricted piece). Add 2: $\boxed{f^{-1}(x) = 2 + \sqrt{x}}$. The restricted f has domain $x \geq 2$ and range $y \geq 0$, so f^{-1} has **domain** $x \geq 0$ and **range** $y \geq 2$.

12. (5.2) f through $(0,1)$, $(1,2)$, $(2,4)$.

- Swap coordinates: f^{-1} goes through $(1,0)$, $(2,1)$, $(4,2)$.
- On the graph, each point flips across the dashed line $y = x$, landing the same distance from the line on the other side. Points on the line itself (where $x = y$) wouldn't move.
- The sketch should look like two mirror images across $y = x$.

13. (5.3) $f(x) = 3 \cdot 2^x$.

- $f(0) = 3 \cdot 1 = \boxed{3}$ (the initial value a).
- $f(3) = 3 \cdot 8 = \boxed{24}$.
- $f(-1) = 3 \cdot \frac{1}{2} = \boxed{1.5}$ (a negative exponent means 1 over).

14. (5.3)

- (a) $500(1.03)^t$: initial value **500**, $b = 1.03 > 1$ so **growth**, $1.03 - 1 = 0.03$, **3%**.
- (b) $80(0.6)^t$: initial value **80**, $b = 0.6 < 1$ so **decay**, $1 - 0.6 = 0.4$, **40%**.

15. (5.3) $f(x) = 2^{x-1} + 3$.

- $x - 1$ inside: shift **right 1**. $+3$ outside: shift **up 3**.
- Domain: **all reals** (exponentials accept any input).
- The asymptote of 2^x is $y = 0$. Shifting up 3 moves it to $\boxed{y = 3}$. The range moves with it: $\boxed{y > 3}$.
- y-intercept: $f(0) = 2^{-1} + 3 = 0.5 + 3 = \boxed{3.5}$.

16. (5.3) $y = ab^x$ through two points.

- (a) $(0,4)$ gives $a = 4$ directly. $(2,36)$: $4b^2 = 36$, so $b^2 = 9$ and $b = 3$. $\boxed{y = 4 \cdot 3^x}$
- (b) $ab = 10$ and $ab^3 = 2.5$. Divide the second by the first: $b^2 = \frac{2.5}{10} = 0.25$, so $b = 0.5$. Then $a(0.5) = 10$, so $a = 20$. $\boxed{y = 20(0.5)^x}$
- Check (b): $20(0.5)^3 = 20 \cdot \frac{1}{8} = 2.5$ ✓.

17. (5.3) Make the bases match, then set the exponents equal.

- (a) $16 = 2^4$: $2^{x+1} = 2^4$, so $x + 1 = 4$ and $\boxed{x = 3}$.
- (b) $9 = 3^2$ and $27 = 3^3$: $(3^2)^x = 3^3$, so $3^{2x} = 3^3$, $2x = 3$, and $\boxed{x = \frac{3}{2}}$.
- (c) $\frac{1}{2} = 2^{-1}$ and $8 = 2^3$: $2^{-x} = 2^3$, so $-x = 3$ and $\boxed{x = -3}$.
- (d) $81 = 3^4$: $3^{x^2} = 3^4$, so $x^2 = 4$ and $\boxed{x = \pm 2}$. (Don't forget the negative root.)

18. (5.3) \$5,000 at 6% for 5 years.

- Quarterly: $n = 4$, so $A = 5000\left(1 + \frac{0.06}{4}\right)^{4 \cdot 5} = 5000(1.015)^{20} \approx 5000(1.34686) \approx \boxed{\$6{,}734.28}$.
- Continuous: $A = 5000e^{0.06 \cdot 5} = 5000e^{0.3} \approx 5000(1.34986) \approx \boxed{\$6{,}749.29}$.
- Continuous compounding earns a little more, as expected.

19. (5.3) 200 g, half-life 10 years.

- $A(t) = 200\left(\frac{1}{2}\right)^{t/10}$ (the exponent $t/10$ counts how many half-lives have passed).
- 25 years is 2.5 half-lives: $A(25) = 200\left(\frac{1}{2}\right)^{2.5} \approx 200(0.17678) \approx \boxed{35.4 \text{ g}}$.
- Sanity check: 2 half-lives leave 50 g and 3 leave 25 g. 35.4 is between ✓.

20. (5.1-5.3) $f(x) = 2^x$, $g(x) = x - 3$.

- (a) $h(x) = f(g(x)) = 2^{x-3}$. That's 2^x shifted **right 3**.
- (b) **Yes.** 2^{x-3} is always increasing (a shifted exponential with $b = 2 > 1$), so it passes the horizontal line test.
- (c) $h(5) = 2^{5-3} = 2^2 = 4$, so $(5, 4)$ is on h. Swap the coordinates: $\boxed{(4, 5)}$ is on h^{-1}.
- (d) h has horizontal asymptote $y = 0$ (shifting right doesn't move it). Reflecting over $y = x$ turns a horizontal line $y = 0$ into the vertical line $\boxed{x = 0}$, which h^{-1} approaches. (Preview: $h^{-1}(x) = \log_2 x + 3$.)